

# AQUÍ VA EL TÍTULO DEL ARTÍCULO

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ABSTRACT. Aquí va el resumen

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## 1. INTRODUCTION

[PP21, SS06a, Hoc69]

## 2. PRELIMINARIES

[Sim06a, Sex03, Joh82]

## 3. COMPACT QUOTIENTS

Necesitamos quitar toda esta sección.

Aquí hay  
que poner la  
referencia.

Let  $f: A \rightarrow B$  be a frame morphism with right adjoint  $f_*: B \rightarrow A$ . If  $F \subseteq A$  and  $G \subseteq B$  are filters, we define:

$$(1) \quad b \in f^*F \iff f_*(b) \in F \quad a \in f_*G \iff f(a) \in G,$$

where  $f^*F$  and  $f_*G$  are filters in  $B$  and  $A$ , respectively.

**Proposition 3.1.** For  $f = f^*: A \rightarrow B$  a frame morphism and  $G \in B^\wedge$ , then  $f_*G \in A^\wedge$ .

*Proof.* By (1),  $f_*G$  is a filter in  $A$ . We need that  $f_*G$  satisfy the open filter condition. Let  $X \subseteq A$  be such that  $\bigvee X \in f_*G$ , with  $X$  directed. Then

$$Y = \{f(x) \mid x \in X\}$$

is directed and  $f(\bigvee X) = \bigvee f[X] = \bigvee Y \in G$ . Since  $G$  is an open filter and  $f(\bigvee X) = \bigvee f[X] \in G$ , there exists  $x \in X$  such that  $f(x) \in G$ . Therefore,  $x \in f_*G$ .  $\square$

Simmons shows in [Sim04, Lemma 8.9 and Corollary 8.10] that the following diagram

$$(2) \quad \begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ U_A \downarrow & & \downarrow U_A \\ \mathcal{O}S & \xrightarrow{F^\infty} & \mathcal{O}S \end{array}$$

commutes laxly, that is,  $U_A \circ f^\infty \leq F^\infty \circ U_A$ . In this diagram  $f^\infty$  and  $F^\infty$  represent the fitted nuclei associated with the filters  $F \in A^\wedge$  and  $\nabla \in \mathcal{O}S^\wedge$ , respectively, that is,

$$f^\infty = v_F \quad \text{and} \quad F^\infty = v_\nabla.$$

Here we prove something more general, since we consider  $j \in NA$ ,  $F \in A_j^\wedge$  and  $j_*F \in A^\wedge$  to produce the diagram

$$\begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ j^* \downarrow & & \downarrow j^* \\ A_j & \xrightarrow{\hat{f}^\infty} & A_j \end{array}$$

where  $\hat{f}^\infty$  and  $f^\infty$  are the fitted nuclei associated with the filters  $j_*F$  and  $F$ , respectively.

The following lemma is part of the analysis developed in this work and constitutes a key tool in the construction of the commutative diagrams that follow.

**Lemma 3.2.** *For  $j = j^*$ ,  $f$  and  $\hat{f}$  as before, it holds that  $j \circ \hat{f} \leq f \circ j$ .*

*Proof.* Recall that, by (1)

$$\hat{f} = \bigvee \{v_y \mid y \in j_*F\} \quad \text{y} \quad f = \bigvee \{v_{j(y)} \mid j(y) \in F\}.$$

Let  $a \in A$  be. For every  $y \in j_*F$ , we have

$$j(v_y(a)) = j(y \succ a) \leq (j(y) \succ j(a)) = v_{j(y)}(j(a)) \leq f(j(a)),$$

where the last inequality follows from  $j(y) \in F$ . Therefore, since  $j$  preserves arbitrary joins,

$$j(\hat{f}(a)) = j\left(\bigvee \{v_y(a) \mid y \in j_*F\}\right) = \bigvee \{j(v_y(a)) \mid y \in j_*F\} \leq f(j(a)).$$

Hence,  $j \circ \hat{f} \leq f \circ j$ .  $\square$

We now prove the above for an arbitrary ordinal.

**Corollary 3.3.** *For  $j$ ,  $f$  and  $\hat{f}$  as before, it holds that  $j \circ \hat{f}^\alpha \leq f^\alpha \circ j$  where  $\alpha \in \text{Ord}$ .*

*Proof.* We proceed by transfinite induction on  $\alpha$ . If  $\alpha = 0$ , the result is immediate.

For the induction step, suppose that the inequality holds for  $\alpha$ . Then

$$j \circ \hat{f}^{\alpha+1} = j \circ \hat{f} \circ \hat{f}^\alpha \leq f \circ j \circ \hat{f}^\alpha \leq f \circ f^\alpha \circ j = f^{\alpha+1} \circ j,$$

where the first inequality follows from Lemma 3.2 and the second from the induction hypothesis.

Finally, let  $\lambda$  be a limit ordinal. For every  $a \in A$ ,

$$j(\hat{f}(a)) = j\left(\bigvee_{\alpha < \lambda} \hat{f}^\alpha\right) = \bigvee_{\alpha < \lambda} j(\hat{f}^\alpha(a)) \leq \bigvee_{\alpha < \lambda} f^\alpha(j(a)) = f^\lambda(j(a)).$$

where the inequality follows from the induction hypothesis. Hence

$$j \circ \hat{f}^\lambda \leq f^\lambda \circ j.$$

This completes the proof.  $\square$

Recalling that  $v_F = f^\infty$  and  $v_{j_*F} = \hat{f}^\infty$ , Corollary 3.3 yields

$$j \circ v_{j_*F} \leq v_F \circ j$$

We now compare the compact quotients induced by  $F$  and  $j_*F$ , and construct a frame morphism between them making the corresponding diagram commute.

If we consider  $j$ ,  $v_F$  and  $v_{j_*F}$  we have the following diagram

$$(3) \quad \begin{array}{ccc} A & \xrightleftharpoons[(v_{j_*F})_*]{(v_{j_*F})^*} & A_{j_*F} \\ \downarrow j & \searrow H & \\ A_j & \xrightleftharpoons[(v_F)^*]{(v_F)_*} & A_F \end{array}$$

where  $A_F$  and  $A_{j_*F}$  are the compact quotients produced by  $v_F$  and  $v_{j_*F}$ , respectively.

Define

$$H: A \rightarrow A_F, \quad H = v_F \circ j.$$

Since  $v_F$  is idempotent,  $H(A) \subseteq A_F$ . Hence, by restricting  $H$  to  $A_{j_*F}$ , we obtain a map

$$h: A_{j_*F} \rightarrow A_F, \quad h = H|_{A_{j_*F}}.$$

As finite meets in quotient frames are inherited from the ambient frame,  $h$  preserves finite meets. It remains to verify that  $h$  preserves arbitrary joins.

**Lemma 3.4.** *The map  $h: A_{j_*F} \rightarrow A_F$  preserves arbitrary joins.*

*Proof.* Let  $X \subseteq A_{j_*F}$ . Since joins in  $A_{j_*F}$  are given by

$$\bigvee_{v_{j_*F}} X = v_{j_*F}(\bigvee X)$$

we have

$$\begin{aligned} h\left(\bigvee_{v_{j_*F}} X\right) &= v_F\left(j\left(v_{j_*F}\left(\bigvee X\right)\right)\right) \\ &\leq v_F\left(v_F\left(j\left(\bigvee X\right)\right)\right) \\ &= v_F\left(j\left(\bigvee X\right)\right), \end{aligned}$$

where the inequality follows from Corollary 3.3 and the equality from the idempotency of  $v_F$ .

Since  $H$  is a frame morphism,

$$v_F\left(j\left(\bigvee X\right)\right) = \bigvee_{v_F}\{H(x) \mid x \in X\} = \bigvee_{v_F}\{h(x) \mid x \in X\}.$$

Therefore,

$$h\left(\bigvee_{v_{j_*F}} X\right) \leq \bigvee_{v_F} h[X].$$

Conversely, for every  $x \in X$ ,

$$x \leq \bigvee_{v_{j_*F}} X,$$

and hence, by monotonicity of  $h$ ,

$$h(x) \leq h\left(\bigvee_{v_{j_*F}} X\right).$$

Since this holds for every  $x \in X$ , we have

$$\bigvee_{v_F} h[X] \leq h\left(\bigvee_{v_{j_*F}} X\right).$$

Thus,

$$h\left(\bigvee_{v_{j_*F}} X\right) = \bigvee_{v_F} h[X].$$

Therefore  $h$  preserves arbitrary joins.  $\square$

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

**Proposition 3.5.** *The diagram*

$$(4) \quad \begin{array}{ccc} A & \xrightarrow{v_{j^*F}} & A_{j^*F} \\ j \downarrow & & \downarrow h \\ A_j & \xrightarrow{v_F} & A_F \end{array}$$

*is commutative.*

*Proof.* By Lemma 3.4,  $h$  is a frame morphism. The commutativity of the diagram follows from the definition of  $h$ .  $\square$

A particular instance of diagram (4) is obtained by taking  $j^* = \mathcal{U}_A$ , where  $S = \text{pt } A$ . Thus, we obtain the following commutative diagram:

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ \mathcal{U}_A \downarrow & & \downarrow h \\ \mathcal{O}S & \xrightarrow{v_\nabla} & \mathcal{O}S_\nabla \end{array}$$

Recall that  $\text{pt } A_F = Q$ . Hence, the spatial reflection of  $A_F$  is given by

$$\mathcal{U}_{A_F}: A_F \rightarrow \mathcal{O} \text{pt } A_F = \mathcal{O}Q.$$

Similarly, since  $\text{pt}((\mathcal{O}S)_\nabla) = Q$ , we have the spatial reflection

$$\mathcal{U}_{\mathcal{O}S_\nabla}: \mathcal{O}S_\nabla \rightarrow \mathcal{O} \text{pt}(\mathcal{O}S)_\nabla = \mathcal{O}Q.$$

Consequently, the two induced morphisms from  $A$  to  $\mathcal{O}Q$  are both given by

$$x \mapsto \mathcal{U}_A(x) \cap Q.$$

Therefore, the preceding constructions fit into the following commutative diagram:

$$(5) \quad \begin{array}{ccccc} A & \xrightarrow{v_F} & A_F & & \\ \mathcal{U}_A \downarrow & & \downarrow h & \searrow \mathcal{U}_{A_F} & \\ & & & & \mathcal{O}Q \\ \mathcal{O}S & \xrightarrow{v_\nabla} & \mathcal{O}S_\nabla & \nearrow \mathcal{U}_{\mathcal{O}S_\nabla} & \end{array}$$

Since the above diagram is determined by  $F \in A^\wedge$  and the associated compact saturated subset  $Q \in \mathcal{Q}S$ , we refer to diagram (5) as the *Q-square*.

**Theorem 3.6.** *Under the hypotheses of the Q-square, if  $A$  satisfies property (H), then*

$$\mathcal{O}Q \cong \uparrow Q'.$$

*Equivalently, the frame of open sets of the point space of  $A_F$  is isomorphic to a compact closed quotient of a Hausdorff space.*

*Proof.* Suppose that  $A$  satisfies **(H)**. Since **(H)** is conservative,  $S = \text{pt } A$  is a  $T_2$  space. Hence, for every  $Q \in \mathcal{Q}S$ , the set  $Q$  is closed, and therefore  $Q' \in \mathcal{O}S$

Moreover, since every  $T_2$  space is strongly stacked, we have

$$v_{\nabla} = [Q'].$$

As  $Q' \in \mathcal{O}S$ , it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_{\nabla}} = (\mathcal{O}S)_{u_{Q'}} \cong \uparrow Q'.$$

Now,  $u_{Q'}$  is a complemented nucleus in  $\mathcal{O}S$ , by [PP12, Proposition VI.3.3], implies that  $\mathcal{O}S_{v_{\nabla}}$  is spatial. Therefore,

$$(\mathcal{O}S)_{v_{\nabla}} \cong \mathcal{O} \text{pt}((\mathcal{O}S)_{v_{\nabla}}).$$

Since

$$\text{pt}((\mathcal{O}S)_{v_{\nabla}}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_{\nabla}} \cong \uparrow Q'.$$

□

#### 4. TIDY FRAMES

**Proposition 4.1.** *If  $A$  is a tidy frame, then  $A_j$  is a tidy frame.*

*Proof.* Let  $A$  be  $\alpha$ -tidy and let  $F \in A^{\wedge}$ . By Proposition 3.1,  $j_*F \in A^{\wedge}$ . Let

$$\hat{f} = \bigvee_{y \in j_*F} v_y \quad \text{and} \quad f = \bigvee_{x \in F} v_x.$$

be the prenuclei associated with  $j_*F$  and  $F$ , respectively.

Since  $A$  is tidy, there exists an ordinal  $\alpha$  such that, for every  $y \in j_*F$ ,

$$y \vee \hat{d} = 1, \quad \text{with } \hat{d} = \hat{f}^{\alpha}(0).$$

Let

$$d = f^{\alpha}(j(0)),$$

by Corollary 3.3,

$$j(\hat{d}) = j(\hat{f}^{\alpha}(0)) \leq f^{\alpha}(j(0)) = d.$$

Since  $j$  is inflationary,  $\hat{d} \leq j(\hat{d})$ , and hence

$$\hat{d} \leq d.$$

Now, let  $x \in F$ . Thus,  $x \in A_j$ , we have  $j(x) = x$ , and therefore  $x \in j_*F$ . Then,

$$x \vee \hat{d} = 1.$$

As  $\hat{d} \leq d$ , it follows that

$$1 = x \vee \hat{d} \leq x \vee d \leq 1.$$

Hence  $x \vee d = 1$ . Therefore,  $A_j$  is  $\alpha$ -tidy.  $\square$

The following theorem establishes that tidiness is preserved under binary coproducts.

**Proposition 4.2.** *Let  $A_1$  and  $A_2$  be  $\alpha$ -tidy frames. Then  $A_1 \oplus A_2$  is  $\alpha$ -tidy.*

*Proof.* By construction of the coproduct  $A_1 \oplus A_2$ , consider the canonical frame morphism

$$\iota_i: A_i \rightarrow A_1 \oplus A_2, \quad i = 1, 2$$

together with their right adjoints  $(\iota_i)_*: A_1 \oplus A_2 \rightarrow A_i$ .

Then, we have the following diagram

$$\begin{array}{ccccccc} & & & \iota_i & & & \\ & & \swarrow & & \searrow & & \\ A_i & \xrightarrow{\alpha_i} & A_1 \times A_2 & \xrightarrow{\downarrow(-)} & \mathcal{L}(A_1 \times A_2) & \xrightarrow{\sim} & A_1 \oplus A_2 \\ & \nwarrow & & & & & \nwarrow \\ & & & (\iota_i)_* & & & \end{array}$$

By Proposition 3.1,

$$(\iota_i)_* F \in A_i^\wedge.$$

Since  $A_i$  is  $\alpha$ -tidy, for every  $x_i \in (\iota_i)_* F$  there exists  $d_i \in A_i$  such that

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

By the definition of  $(\iota_i)_* F$ ,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Thus,

$$x_1 \oplus 1, 1 \oplus x_2 \in F.$$

Since  $F$  is a filter,

$$(x_1 \oplus 1) \wedge (1 \oplus x_2) = x_1 \oplus x_2 \in F.$$

Now let

$$f_F = \bigvee_{z \in F} v_z$$

be the prenucleus associated with  $F$ . We first show that, for  $i = 1, 2$ ,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

Indeed, let  $a \in A_i$  and  $y \in (\iota_i)_* F$ . Since  $\iota_i(y) \in F$ ,

$$\begin{aligned} \iota_i(v_y(a)) &= \iota_i(y \succ a) \\ &\leq (\iota_i(y) \succ \iota_i(a)) \\ &= v_{\iota_i(y)}(\iota_i(a)) \\ &\leq f_F(\iota_i(a)). \end{aligned}$$

Therefore, since the join defining  $f_i$  is pointwise and  $\iota_i$  preserves arbitrary joins,

$$\iota_i(f_i(a)) = \iota_i\left(\bigvee_{y \in (\iota_i)_* F} v_y(a)\right) = \bigvee_{y \in (\iota_i)_* F} \iota_i(v_y(a)) \leq f_F(\iota_i(a)).$$

Hence,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

We now prove by transfinite induction that

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal  $\beta$ .

If  $\beta = 0$ , the result is immediate. Suppose that it holds for  $\beta$ . Then

$$\begin{aligned} \iota_i \circ f_i^{\beta+1} &= \iota_i \circ f_i \circ f_i^\beta \\ &\leq f_F \circ \iota_i \circ f_i^\beta \\ &\leq f_F \circ f_F^\beta \circ \iota_i \\ &= f_F^{\beta+1} \circ \iota_i. \end{aligned}$$

Now let  $\lambda$  be a limit ordinal. For every  $a \in A_i$ ,

$$\begin{aligned} \iota_i(f_i^\lambda(a)) &= \iota_i\left(\bigvee_{\beta < \lambda} f_i^\beta(a)\right) \\ &= \bigvee_{\beta < \lambda} \iota_i(f_i^\beta(a)) \\ &\leq \bigvee_{\beta < \lambda} f_F^\beta(\iota_i(a)) \\ &= f_F^\lambda(\iota_i(a)). \end{aligned}$$

Thus,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal  $\beta$ .

Evaluating at  $0 \in A_i$  for  $\beta = \alpha$ , we obtain

$$\iota_1(d_1) \leq f_F^\alpha(0 \oplus 1) \quad \text{and} \quad \iota_2(d_2) \leq f_F^\alpha(1 \oplus 0).$$

Therefore,

$$\begin{aligned} d_1 \oplus d_2 &= (d_1 \oplus 1) \wedge (1 \oplus d_2) \\ &\leq f_F^\alpha(0 \oplus 1) \wedge f_F^\alpha(1 \oplus 0) \\ &= f_F^\alpha(0 \oplus 1 \wedge 1 \oplus 0). \end{aligned}$$

Set  $d = f_F^\alpha(0 \oplus 0)$ , then  $d_1 \oplus d_2 \leq d$ . Finally,

$$\begin{aligned} 1 \oplus 1 &= (x_1 \vee d_1) \oplus (x_2 \vee d_2) \\ &= (x_1 \oplus x_2) \vee (d_1 \oplus d_2) \\ &\leq (x_1 \oplus x_2) \vee d \\ &\leq 1 \oplus 1. \end{aligned}$$

Therefore,

$$(x_q \oplus x_2) \vee d = 1 \oplus 1$$

and  $A_1 \oplus A_2$  is  $\alpha$ -tidy.  $\square$

**Corollary 4.3.** *Let  $A_1$  and  $A_2$  be  $\alpha$ -tidy and  $\beta$ -tidy, respectively. Then  $A_1 \oplus A_2$  is  $\gamma$ -tidy, where*

$$\gamma = \max\{\alpha, \beta\}.$$

*Proof.* Let  $\gamma = \max\{\alpha, \beta\}$ . Then  $A_1$  and  $A_2$  are both  $\gamma$ -tidy. Hence, by Proposition 4.2,  $A_1 \oplus A_2$  is  $\gamma$ -tidy.  $\square$

We extend the previous result to arbitrary coproducts.

**Proposition 4.4.** *Tidiness is preserved under arbitrary coproducts.*

*Proof.* Let  $\{A_i\}_{i \in I}$  be a family of tidy frames, and for each  $i \in I$ , let  $\alpha_i$  be a tidiness degree of  $A_i$ . Consider

$$A = \bigoplus_{i \in I} A_i$$

and let

$$\iota_i: A_i \rightarrow A$$

denote the canonical coproduct morphisms. Since  $\iota_i$  is a frame morphism, it has a right adjoint  $(\iota_i)_*$ .

Let  $F \in A^\wedge$ . By Proposition 3.1,

$$(\iota_i)_* F \in A_i^\wedge$$

for every  $i \in I$ . Moreover,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Set  $\alpha = \sup\{\alpha_i \mid i \in I\}$ . Then every  $A_i$  is  $\alpha$ -tidy. Hence, for every  $x_i \in (\iota_i)_* F$ ,

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

Let

$$f_F = \bigvee_x \cdot \in Fv_x$$

be the prenucleus associated with  $F$ . As in the proof of Proposition 4.2, for every  $i \in I$  we have

$$\iota_i \circ f_i \leq f_F \circ \iota_i,$$

and, by transfinite induction,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal  $\beta$ .

Evaluating at  $0 \in A_i$  for  $\beta = \alpha$ , we obtain

$$\iota_i(d_i) = \iota_i(f_i^\alpha(0)) \leq f_F^\alpha(\iota_i(0)) = f_F^\alpha(\langle 0 \rangle).$$

Set

$$d = f_F^\alpha(\langle 0 \rangle).$$

Therefore,

$$\iota_i(d_i) \leq d$$

for every  $i \in I$ . Taking arbitrary joins yields

$$\langle d_i \rangle \bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let

$$\langle x_i \rangle \in F.$$

Since  $x_i \vee d_i = 1$  for every  $i \in I$ , we have

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle.$$

As  $\langle d_i \rangle \leq d$ ,

$$\langle 1 \rangle = \langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d \leq \langle 1 \rangle.$$

Thus,

$$\langle x_i \rangle \vee d = \langle 1 \rangle.$$

Therefore,  $A = \bigoplus_{i \in I} A_i$  is  $\alpha$ -tidy, and hence tidy.  $\square$

**Corollary 4.5.** *The subcategory of tidy frames is a coreflective subcategory of  $\mathbf{Frm}$ .*

## 5. STACKED AND STRONGLY STACKED FRAMES

**Lemma 5.1.** *If  $\alpha \in \mathbf{Ord}$  and  $X \in CS$ , then*

$$(6) \quad \partial_F^\alpha(X) = (f_F^\alpha(X'))'.$$

*Proof.* By transfinite induction on  $\alpha \in \mathbf{Ord}$ :

- If  $\alpha = 0$ , then  $\partial_F^0 = \text{id}$ , that is,  $\partial_F^0(X) = X$  and  $f_F^0 = \text{id}$ . Thus

$$\partial_F^0(X) = X = (X'') = (f_F^0(X'))'.$$

- Assume that  $\partial_F^\alpha(X) = (f_F^\alpha(X'))'$  holds. By definition,

$$\partial_F^{\alpha+1}(X) = \partial_F(\partial_F^\alpha(X)).$$

Furthermore,

$$\partial_F^{\alpha+1}(X) = (f_F((\partial_F^\alpha(X))'))'.$$

By induction hypothesis,  $\partial_F^\alpha(X) = (f_F^\alpha(X'))'$  and substituting we obtain

$$\partial_F^{\alpha+1}(X) = (f_F(f_F^\alpha(X')))' = (f_F^{\alpha+1}(X'))'$$

which is what we wanted.

- Let  $\lambda$  be a limit ordinal and assume that the statement holds for all  $\alpha < \lambda$ . By definition,

$$\partial_F^\lambda(X) = \bigcap_{\alpha < \lambda} \partial_F^\alpha(X).$$

By De Morgan laws, we have that

$$(\partial_F^\lambda(X))' = \left( \bigcap_{\alpha < \lambda} \partial_F^\alpha(X) \right)' = \bigcup_{\alpha < \lambda} (\partial_F^\alpha(X))'.$$

Using the induction hypothesis, for all  $\alpha < \lambda$ ,

$$(\partial_F^\alpha(X))' = f_F^\alpha(X'),$$

Then,

$$(\partial_F^\lambda(X))' = \bigcup_{\alpha < \lambda} f_F^\alpha(X') = f_F^\lambda(X').$$

Therefore,

$$(\partial_F^\lambda(X))' = f_F^\lambda(X'),$$

that is,

$$\partial_F^\lambda(X) = (f_F^\lambda(X'))'.$$

□

As a particular case, we have the following.

**Corollary 5.2.** *If  $X \in \mathcal{CS}$ ,  $U \in \mathcal{OS}$  and  $\alpha = \infty$ , then*

$$(7) \quad \begin{aligned} \partial_F^\infty(X) = (v_F(X'))' &\iff \partial_F^\infty(U')' = (v_F(U)) \\ &\iff \partial_F^\infty(X)' = (v_F(X')). \end{aligned}$$

**Lemma 5.3.** *If  $A = \mathcal{OS}$  and  $E \subseteq S$ , then*

$$\mathcal{U}_*[E]\mathcal{U} = [E].$$

*Proof.* Since  $A$  is spacial, then  $\mathcal{U}$  is an isomorphism. Thus, if  $U \in \mathcal{O} \text{ pt } A$  exists  $V \in \text{pt } A$  such that  $U = \mathcal{U}(V)$  and as  $U \in \text{pt } A$ , then  $U = V$ . Next,

$$\begin{aligned} (\mathcal{U}_*[E]\mathcal{U})(U) &= \mathcal{U}_*[E](\mathcal{U}(U)) = \mathcal{U}_*([E](U)) \\ &= \mathcal{U}_*(E \cup U)^\circ = \bigcup \{W \in \mathcal{OS} \mid W \subseteq (E \cup U)^\circ\} \\ &= (E \cup U)^\circ = [E](U). \end{aligned}$$

Therefore  $\mathcal{U}_*[E]\mathcal{U} = [E]$ .

□

**Proposition 5.4.** . For  $F \in A^\wedge$ ,  $Q \in \mathcal{QS}$  and  $\nabla = \nabla(Q)$ , if  $k \in [v_\nabla, w_\nabla]$ , then  $\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F]$ .

*Proof.* If  $k \in [v_\nabla, w_\nabla]$ , then  $\mathcal{U}_* \circ k \circ \mathcal{U} \in NA$ . Additionally, for  $F \in A^\wedge$  if  $j \in [v_F, w_F]$ , then  $F = \nabla(j)$ . In this way we must prove that  $F = \nabla(\mathcal{U}_* \circ k \circ \mathcal{U})$ .

By Hofmann-Mislove Theorem, we have that

$$x \in F \iff Q \subseteq \mathcal{U}(x) \iff \mathcal{U}(x) \in \nabla = \nabla(Q).$$

Furthermore, for all  $x \in A$

$$(\mathcal{U}_* \circ k \circ \mathcal{U})(x) = \mathcal{U}_*(k(\mathcal{U}(x))) = \bigwedge (S \setminus k(\mathcal{U}(x))).$$

Since  $k \in [v_\nabla, w_\nabla]$ , then  $\nabla = \nabla(k)$ , that is, for  $V \in \nabla$  it holds that  $k(V) = S$ . Therefore,

$$\begin{aligned} x \in F &\iff \mathcal{U}(x) \in \nabla \iff S \setminus k(\mathcal{U}(x)) = \emptyset \\ &\iff \bigwedge (S \setminus k(\mathcal{U}(x))) = 1 \\ &\iff x \in \nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) \end{aligned}$$

□

We have a morphism  $\mathcal{U}: [v_\nabla, w_\nabla] \rightarrow [v_F, w_F]$ . Note that this occurs at both ends of the interval on  $NA$ .

Take  $k \in [v_\nabla, w_\nabla]$ . Since  $v_F$  is a fitted nuclei, then it holds

$$(8) \quad v_F \leq \mathcal{U}_* \circ k \circ \mathcal{U}.$$

Similarly, as  $w_F$  is the largest element in its block,

$$(9) \quad \mathcal{U}_* \circ k \circ \mathcal{U} \leq w_F.$$

**Proposition 5.5.** If  $F \in A^\wedge$  and  $\nabla \in \mathcal{OS}^\wedge$ , then

$$w_F = \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}.$$

*Proof.* By (9), we must verify that  $w_F \leq \mathcal{U}_* \circ w_\nabla \circ \mathcal{U}$ . Equivalently, we must prove that

$$\mathcal{U} \circ w_F \leq w_\nabla \circ \mathcal{U},$$

that is, for all  $x \in A$

$$\mathcal{U}(w_F(x)) \subseteq [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Since  $\mathcal{U}(w_F(x)) \in \mathcal{OS}$ , it is enough that  $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$ . As a contradiction, let  $y \in \mathcal{U}(w_F(x))$  be such that  $y \notin M'$  and  $y \notin \mathcal{U}(x)$ . Then

$$y \in M \quad \text{and} \quad x \leq y.$$

By hypothesis,  $y \in \mathcal{U}(w_F(x))$ , then

$$w_F(x) \not\leq y \quad \text{and} \quad w_F(x) = \bigwedge \{m \in M \mid x \leq m\},$$

that is,  $w_F(x) \leq y$  and it is a contradiction.

Therefore,  $\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x)$ .  $\square$

**Definition 5.6.** Let  $A$  be a frame.

- (1)  $A$  is *strongly stacked* if  $\mathcal{U}_*[Q']\mathcal{U} = v_F$ .
- (2)  $A$  is *stacked* if  $(\mathcal{U}_*([Q'])\mathcal{U})(0) = v_F(0)$ .

for any  $(F, Q)$  HM pair.

Note that strongly stacked implies stacked. Our aim is to prove that the above properties are conservative, that is,  $S$  is a stacked (strongly stacked) space if and only if its frame of open sets  $\mathcal{O}S$  is stacked (strongly stacked).

**Proposition 5.7.** *The notions of strongly stacked and stacked are conservative.*

*Proof.* Consider  $S \in \text{Top}$ ,  $A = \mathcal{O}S$ ,  $F \in A^\wedge$  and  $Q \in \mathcal{Q}S$ .

First, suppose that  $S$  is strongly stacked, then  $v_F = [Q']$  and, by Lemma 5.3,  $\mathcal{U}_*[Q']\mathcal{U} = [Q'] = v_F$ . Therefore  $\mathcal{O}S$  is strongly stacked.

Now, suppose that  $S$  is stacked, then

$$\overline{Q} = \partial_F^\infty(S) \iff \overline{Q}' = (\partial_F^\infty(S))'.$$

Furthermore,

$$\begin{aligned} (\mathcal{U}_*[Q']\mathcal{U})(\emptyset) &= \mathcal{U}_*([Q'](\{V \in \text{pt } \mathcal{O}S \mid \emptyset \not\subseteq V\})) \\ &= \mathcal{U}_*([Q'](\emptyset)) = \mathcal{U}_*(Q')^\circ \\ &= \bigcup \{W \in \mathcal{O}S \mid W \subseteq (Q')^\circ\} \\ &= (Q')^\circ = \overline{Q}'. \end{aligned}$$

By Corollary 5.2,  $v_F(\emptyset) = (\partial_F^\infty(\emptyset))'$ . Therefore,

$$v_F(\emptyset) = (\mathcal{U}_*[Q']\mathcal{U})(\emptyset),$$

and  $\mathcal{O}S$  is stacked.  $\square$

## 6. SEPARATION PROPERTIES IN FRAMES

If  $S \in \text{Top}$  is an  $T_1$  and strongly stacked, then it holds that  $w_\nabla = v_\nabla$ . Additionally, if  $S$  is stacked and  $T_1$ , then it is sober.

Thus, we could ask ourselves if the same occurs in frames. In this section we will study the above question.

Take  $p \in \text{pt } A$  and  $(F_p, \uparrow p)$  a HM pair, then for

$$M = \{m \in A \setminus F_p \mid (\forall a \in A \setminus F_p)[a \leq m]\},$$

the nuclei  $w_{F_p}(x) = \bigwedge \{m \in M \mid x \leq m\}$  is  $w_p(x)$ .

To prove this, note that  $A \setminus F_p = \{x \in A \mid x \leq p\} = \downarrow p$ . As,

$$M = \{m \in \downarrow p \mid [\forall a \in \downarrow p](a \leq m)\} = \{p\}.$$

Hence,  $w_{F_p}(x) = \bigwedge \{p \mid x \leq p\}$ , that is,

$$(10) \quad w_{F_p}(x) = \begin{cases} 1, & \text{if } x \not\leq p \\ p, & \text{if } x \leq p \end{cases} = w_p(x)$$

for all  $x \in A$ . Therefore  $w_{F_p} = w_p$ .

**Proposition 6.1.** *Let  $A \in \text{Frm}$  and  $(F, Q)$  be any HM pair. If  $v_F(0) = w_F(0)$ , then for all  $p \in \text{pt } A$ ,  $p$  is maximal in  $\text{pt } A$ .*

*Proof.* Consider  $p, q \in \text{pt } A$  such that  $p \leq q$ . Note that for  $q \in \text{pt } A$ , we have  $(F_q, \uparrow q)$  a HM pair, with

$$F_q = \{x \in A \mid \uparrow q \subseteq \mathcal{U}(x)\} = \{x \in A \mid (\forall r \in \text{pt } A)[r \leq q \Rightarrow x \not\leq r]\}.$$

By hypothesis,  $v_{F_q}(0) = w_{F_q}(0)$  and by (10)

$$q = w_q(0) = w_{F_q}(0) = v_{F_q}(0) \leq v_{F_q}(p) = p.$$

Therefore,  $p = q$  and every point is maximal in  $\text{pt } A$ .  $\square$

As we show later on (2), for all  $(F, Q, \nabla)$  HM tupla we have

$$\mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U}.$$

Additionally, if  $A$  is strongly stacked, then  $v_F = \mathcal{U}_*[Q']\mathcal{U}$  and

$$\mathcal{U}_* \circ v_\nabla \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q']\mathcal{U}$$

Thus,  $v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}$ , equivalently  $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ .

Therefore, if  $A$  is strongly stacked, then the diagram (2) is commutative. The following result tells us under which conditions we have the converse implication.

**Proposition 6.2.** *If  $A \in \text{Frm}$  is tidy and the diagram*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A \\ \mathcal{U} \downarrow & & \downarrow \mathcal{U} \\ \mathcal{OS} & \xrightarrow{v_\nabla} & \mathcal{OS} \end{array}$$

*commutes, then  $A$  is strongly stacked.*

*Proof.* Let  $A \in \text{Frm}$ ,  $S = \text{pt } A$  and  $(F, Q)$  be any HM pair. By hypothesis,  $A$  is tidy and by previous results we have the implications

$$A \text{ tidy} \Rightarrow \mathcal{OS} \text{ tidy} \Rightarrow S \text{ stacked and packed}.$$

Since  $S$  is packed, if  $Q \in \mathcal{QS}$ , then  $Q \in \mathcal{CS}$  and  $Q' \in \mathcal{OS}$ . Then, as  $S$  is stacked, we have that  $\overline{Q} = \partial_\nabla^\infty(S)$ , that is,  $Q' = v_\nabla(\emptyset)$  and as  $\mathcal{OS}$  is tidy,  $v_\nabla = u_{Q'} = [Q']$ .

Additionally, if  $v_\nabla \circ \mathcal{U} = \mathcal{U} \circ v_F$ , then

$$v_\nabla \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{and} \quad \mathcal{U} \circ v_F \leq v_\nabla \circ \mathcal{U},$$

that is,

$$[Q'] \circ \mathcal{U} \leq \mathcal{U} \circ v_F \quad \text{and} \quad \mathcal{U} \circ v_F \leq [Q'] \circ \mathcal{U}.$$

By properties of the right adjoint

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq v_F \quad \text{and} \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore  $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$  and  $A$  is strongly stacked.  $\square$

Consequently, we have the following result.

**Corollary 6.3.** *If  $A$  is tidy, then  $S = \text{pt } A$  is strongly stacked.*

*Proof.* Since  $S$  is strongly stacked if and only if  $v_\nabla = [Q']$ , the result follows from the fact that

$$\mathcal{O}S \text{ tidy} \Rightarrow S \text{ strongly stacked}$$

$\square$

With the morphism  $\mathcal{U}_A$ , we have the nuclei  $\text{sp}_A$  defined by

$$\text{sp}_A(x) = \bigwedge \{p \in \text{pt } A \mid x \leq p\}$$

for  $x \in A$ . Combining this fact with the Proposition 6.1 we know information about the collapse on admissibility interval.

**Proposition 6.4.** *For  $A \in \text{Frm}$  and  $(F, Q)$  any HM pair, the following statements hold:*

- (1) *If  $A$  is  $T_1$  and stacked, then  $v_F(0) = w_F(0)$ .*
- (2) *If  $v_F(0) = w_F(0)$ , then  $A$  is stacked.*
- (3) *If  $v_F(0) = w_F(0)$  and the set  $Y = \{q \in \text{pt } A \mid a \leq q\}$  is non-empty for all  $a \in A$ , then  $A$  is  $T_1$ .*

*Proof.*

- (1) If  $A$  is  $T_1$ , then

$$\mathcal{O}S \text{ is } T_1 \Rightarrow S \text{ is } T_1 \Rightarrow Q = M.$$

Since  $A$  is stacked,

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) = w_F(0).$$

- (2) Recall that for every HM pair, the following inequality holds:

$$(11) \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F$$

and by hypothesis  $v_F(0) = w_F(0)$ , that is,  $v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = w_F(0)$ . Therefore,  $A$  is stacked.

- (3) Consider  $p \in \text{pt } A$  and  $a \in A$  such that  $p \leq a$ . By hypothesis  $\{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$ . Then

$$p \leq a \leq \text{sp}(a) = \bigwedge \{q \in \text{pt } A \mid a \leq q\}$$

for some  $q \in Y$ . Thus,  $p \leq a \leq q$  and by Proposición 6.1,  $p = q$ . Therefore,  $p = a$  and every point is maximal in  $A$ , that is,  $A$  is  $T_1$ .  $\square$

**Corollary 6.5.** *For  $A \in \text{Frm}$  and  $(F, Q)$  any HM pair, the following statements hold:*

- (1) *If  $A$  is  $T_1$  and strongly stacked, then  $v_F = w_F$ .*
- (2) *If  $v_F = w_F$ , then  $A$  is strongly stacked.*
- (3) *If  $v_F = w_F$  and the set  $Y = \{q \in \text{pt } A \mid a \leq q\}$  is non-empty for all  $a \in A$ , then  $A$  is  $T_1$ .*

*Proof.*

- (1) If  $A$  is strongly stacked, then  $v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$  and by  $T_1$  we have that  $Q = M$ . Therefore  $v_F = w_F$ .
- (2) This follows from (11).
- (3) This follows from Proposition 6.4.

□

Note that, by 6.1 and 6.4, if  $v_F(0) = w_F(0)$ , then  $A$  is stacked and every point is maximal in  $\text{pt } A$ . Particularly, for each  $p \in \text{pt } A$  and  $(F_p, \uparrow p)$  it holds  $v_{F_p}(0) = p$ . Furthermore,

$$v_{F_p}(p) = v_{F_p}(0) = 0_{A_{F_p}}.$$

If  $A$  is  $T_1$ ,  $p$  is maximal in  $A$ , then  $p$  is maximal in  $A_{F_p}$  and  $A_{F_p} = \{q, 1\}$ .

Now,  $F_{F_p} = \nabla(v_{F_p})$  and  $x \in F_p$  if and only if  $v_{F_p}(x) = 1$ . In this way, under the previous hypotheses, it holds that

$$v_{F_p}(x) = \begin{cases} 1, & \text{si } x \not\leq p \\ p, & \text{si } x \leq p \end{cases} = \begin{cases} 1, & \text{si } x \in F_p \\ p, & \text{si } x \notin F_p \end{cases} = w_{F_p}(x).$$

Therefore, if  $A$  is  $T_1$  and stacked, then  $v_{F_p} = w_{F_p}$  for each  $p \in \text{pt } A$ . With this in mind, we could ask ourselves if there exists some way to recover the notion of strongly stacked by means of  $T_1$  and stacked.

Now, we need the following information:

- (1) Let  $A^\wedge$  be a topological space where the basic open sets of  $\mathcal{O}(A^\wedge)$  are given by

$$D(x) = \{F \in A^\wedge \mid x \in F\}$$

where  $x \in A$ .

- (2) For  $A \in \text{Frm}$ , we can construct a frame  $VA$  as in [Sim04]. Each element in  $\text{pt}(VA)$  is of the form

$$(F, a), \quad \text{where } F \in A^\wedge \text{ y } a \in [v_F(0), w_F(0)].$$

- (3) Take a continuous surjective map  $\pi: \text{pt}(VA) \rightarrow A^\wedge$ , defined by

$$(F, a) \mapsto F.$$

- (4) By the functor  $\mathcal{O}(-)$  we have

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \rightarrow \mathcal{O}(\text{pt}(VA)).$$

(5) Open sets in  $\text{pt}(VA)$  are given by the following:

$$(12) \quad (F, a) \in [-](x) \iff x \in F \quad \text{and} \quad (F, a) \in \langle - \rangle(x) \iff x \not\leq a.$$

Particularly,

$$(13) \quad a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F \rangle(x) \iff x \not\leq a.$$

**Proposition 6.6.** *Let  $A, VA \in \text{Frm}$  and  $A^\wedge$  be as above. The following statements hold:*

- (1) *If  $A$  is  $T_1$  and stacked, then  $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$  under the continuous function  $\pi$ .*
- (2) *If the morphism  $\mathcal{O}(\pi) = \pi^{-1}$  induces that  $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$ , then  $A$  is stacked.*
- (3) *If the morphism  $\mathcal{O}(\pi) = \pi^{-1}$  induces that  $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$  and*

$$\mathcal{V}(a) = \{q \in \text{pt } A \mid a \leq q\} \neq \emptyset,$$

*then  $A$  is  $T_1$ .*

*Proof.*

- (1) If  $A$  is  $T_1$  and stacked, then for each  $F \in A^\wedge$  it holds that

$$v_F(0) = w_F(0).$$

Take  $(F, a), (G, b) \in \text{pt}(VA)$  such that  $\pi(F, a) = F = \pi(G, a) = G$ . By  $T_1$  and stacked we have

$$a = v_F(0) = v_G(0) = b.$$

Therefore  $(F, a) = (G, b)$ , that is,  $\pi$  is injective. Since it is also surjective,  $\pi$  is bijective.

We must show that  $\pi^{-1}: A^\wedge \rightarrow \text{pt}(VA)$  is continuous. For this, recall that by (12)

$$\begin{aligned} (F, a) \in [-](x) &\iff x \in F \\ &\iff (F, a) \in \pi^{-1}(D(x)) \\ &\iff \pi([-](x)) = D(x). \end{aligned}$$

For the other basic open sets, by (13) we have

$$a \notin \langle F \rangle(x) \iff x \leq a \iff v_F(x) \leq a.$$

By  $T_1$  and stacked we have that  $a = v_F(0)$  and thus

$$v_F(0) \notin \langle F \rangle(x) \iff x \leq v_F(0) \iff v_F(x) = v_F(0).$$

As  $v_F(0) \neq 1$ ,

$$v_F(0) \notin \langle F \rangle(x) \iff x \notin F$$

or equivalently

$$a \in \langle F \rangle(x) \iff (F, a) \in \langle - \rangle(x) \iff x \in F.$$

Therefore,  $\pi(\langle - \rangle(x)) = D(x)$  y  $\pi^{-1}$  is continuous. Hence,  $\pi$  is a homeomorphism and

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA)).$$

- (2) Suppose that  $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$  under  $\mathcal{O}(\pi) = \pi^{-1}$ . Since  $\pi^{-1}$  is bijective, then  $\pi$  is also bijective. Thus, if  $F \in A^\wedge$ , then

$$(F, v_F(0)), (F, w_F(0)) \in \text{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

By the injectivity of  $\pi$  we have that  $v_F(0) = w_F(0)$  and therefore  $A$  is stacked.

- (3) Suppose that  $\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA))$  and  $A$  is stacked. Consider  $a \in A$  and  $p \in \text{pt } A$  such that  $p \leq a$ . Moreover, by the previous item we have that

$$\mathcal{O}(A^\wedge) \cong \mathcal{O}(\text{pt}(VA)) \Rightarrow v_F(0) = w_F(0)$$

and every point is maximal in  $\text{pt } A$ . Since  $\mathcal{V}(a) \neq \emptyset$ , there exists  $q \in \text{pt } A$  such that  $a \leq q$ . By the maximality of  $p$  and  $q$ , we have

$$p \leq a \leq q = p.$$

Therefore,  $p = a$  and  $A$  is  $T_1$ .

□

Note that item (3) holds whenever  $A$  is spatial. In the spatial setting, this corresponds to the topological fact that

$$S \text{ is } T_1 \text{ and stacked} \Rightarrow S \text{ is sober.}$$

However, the point-free setting allows for a potentially broader phenomenon, since the conditions of being  $T_1$  and stacked do not a priori force a frame to be spatial. This leads naturally to the following question:

Do there exist  $T_1$  stacked frames that are not spatial?

## 7. QUOTIENTS OF STACKED FRAMES

Remember that for  $(f: A \rightarrow B) \in \text{Frm}$ ,  $j \in NA$  and  $k \in NB$ , we have the following definitions:

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NA_j$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where  $f_*$  is the right adjoint of  $f$ .

**Proposition 7.1.** *Let  $A$  be a frame and  $j \in NA$ . If  $A$  is strongly stacked, then  $A_j$  is strongly stacked.*

*Proof.* Consider  $S = \text{pt } A$ ,  $S_j = \text{pt } A_j$  and let  $F \in A_j^\wedge$  be. We define:

$$G = j_*(F), \quad Q = S_j \setminus F, \quad R = S \setminus G,$$

where  $j_*$  is the right adjoint associated with the frame morphism  $j^*: A \rightarrow A_j$  and  $x \in G$  if and only if  $j(x) \in F$ . Considering the spatial reflections

$$\mathcal{U}: A \rightarrow \mathcal{O}S \quad \text{and} \quad \mathcal{U}_j: A_j \rightarrow \mathcal{O}S_j$$

we have the following diagram

$$\begin{array}{ccccc}
 & & A & \xrightarrow{\mathcal{U}} & \mathcal{O}S \\
 & \nearrow v_G & \downarrow j^* & \nearrow [R'] & \downarrow sj \\
 A & \xrightarrow{\mathcal{U}} & \mathcal{O}S & & \\
 \downarrow j^* & & \downarrow sj & \nearrow v_F & \\
 A_j & \xrightarrow{\mathcal{U}_j} & \mathcal{O}S_j & \xrightarrow{\mathcal{U}_j} & \mathcal{O}S_j \\
 & \nearrow [Q'] & & & 
 \end{array}$$

Using the right adjoints of  $\mathcal{U}$  and  $\mathcal{U}_j$  we obtain

$$v_F, \quad l = N(\mathcal{U}_j)_*([Q']), \quad m = N(j^*)(N(\mathcal{U})_*([R'])), \quad n = N(j^*)(N(\mathcal{U}_j \circ j^*)_*)([Q'])$$

are nuclei in  $NA_j$ . Let us see what is the relationship between these nuclei.

By construction, we have  $v_F \leq N(\mathcal{U}_j)_*([Q'])$  and we must verify the other inequality for  $A_j$  to be strongly stacked.

By hypothesis, we have  $v_G = N(\mathcal{U})_*([R'])$ . Moreover, for  $x \in A$ , we have  $j(v_x) \leq v_{j(x)}$  and, by definition of the adjusted nuclei associated with  $F$  and  $G$ ,

$$(14) \quad j \circ v_G \leq v_F.$$

Furthermore, for  $(j: A \rightarrow A_j) \in \text{Frm}$

$$N(j^*)(v_G) = \bigvee \{u_{j(v_G(a))} \wedge v_{j(a)} \mid a \in A\} \quad \text{y} \quad v_F = \bigvee \{u_{v_F(a)} \wedge v_a \mid a \in A_j\}.$$

Note, by (14),  $u_{j(v_G)} \leq u_{v_F}$  and therefore

$$(15) \quad m \leq v_F.$$

To  $l$  and  $n$ , we have

$$n = \bigvee \{u_{j^*(n(x))} \wedge v_{j^*(x)} \mid x \in A\} \quad \text{and} \quad l = \bigvee \{u_{l(x)} \wedge v_x \mid x \in A_j\},$$

where

$$j^*(n(x)) = j^*((j_* \circ \mathcal{U}_{j*} \circ [Q'] \circ \mathcal{U}_j \circ j^*)(x)) \quad \text{y} \quad l(x) = N(\mathcal{U}_j)_*([Q'])(x).$$

Then,

$$\begin{aligned} j^*(n(x)) &= j^*(j_*(\bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\})) \\ &= \bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}. \end{aligned}$$

Since  $x \in A_j$ ,

$$l(x) = \bigvee\{a \in A \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j(x))\}.$$

Therefore  $j(n(x)) = l(x)$ .

To  $m(x)$  we have

$$m(x) = \bigvee\{u_{j^*}(\mathcal{U}_* \circ [R'] \circ \mathcal{U})(x) \wedge v_{j^*}(x) \mid x \in A\}.$$

Since  $A_j \subseteq A$ , then  $Q = R \cap A_j$  and

$$m(x) = \bigvee\{u_{j^*}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x) \wedge v_x \mid x \in A_j\}.$$

Note

$$\begin{aligned} j^*(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(x) &= j^*(\bigvee\{a \in A \mid \mathcal{U}(a) \subseteq Q' \cup \mathcal{U}(x)\}) \\ &= \bigvee\{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\}. \end{aligned}$$

To  $x \in A_j$

$$\begin{aligned} \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(j^*(x)) &\iff \mathcal{U}(j^*(a)) \cap S_j \subseteq (Q' \cup \mathcal{U}(j^*(x))) \cap S_j \\ &\iff \mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(j^*(x)), \end{aligned}$$

that is,

$$\bigvee\{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\} = j^*(N(\mathcal{U}_j \circ j)_*)([Q']).$$

Thus, we have

$$\bigvee\{u_{j^*}(N(\mathcal{U}_j \circ j)_*)([Q'])(x) \wedge x \mid x \in A\} = m(x).$$

Since  $(N(\mathcal{U}_j \circ j)_*)([Q']) \leq j(N(\mathcal{U}_j \circ j)_*)([Q'])$  and by (15)

$$l \leq m \leq v_F \leq l.$$

Therefore  $v_F = \mathcal{U}_{j^*} \circ [Q'] \circ \mathcal{U}_j$ , that is,  $A_j$  is strongly stacked.  $\square$

Since every strongly stacked frame is stacked, we obtain the following result.

**Corollary 7.2.** *Let  $A$  be a frame and  $j \in NA$ . If  $A$  is strongly stacked, then  $A_j$  is stacked.*

*Proof.* By the previous proposition, if  $A$  is strongly stacked, then  $A_j$  is strongly stacked and this implies that  $A_j$  is stacked.  $\square$

## 8. RELATIONSHIP TO THE PROPERTY (H)

If  $A$  is a Hausdorff frame, then  $A$  is  $T_1$ . In this way, to be able to use the example of Paseka and Smarda it is enough that (H) implies stacked.

**Proposition 8.1.** *If  $A \in \text{Frm}$  satisfies (H), then  $A$  is stacked.*

*Proof.* If  $A$  satisfies (H), then  $A$  is  $T_1$ , that is, for every pair of points  $x, y \in A$  with  $x \neq y$ , there exist neighborhoods  $U, V$  of  $x, y$  respectively such that  $U \cap V = \emptyset$ . This implies that

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F,$$

that is,  $\mathcal{U}_* \circ [Q'] \circ \mathcal{U}(0) = w_F(0)$  and since  $w_F \neq \text{tp}$ , we have  $w_F(0) \neq 1$ .

By construction, we have that  $v_F(0) \leq w_F(0)$  and thus, we need to verify that the reverse inequality holds to conclude that  $A$  is stacked. For a contradiction, suppose that

$$w_F(0) \not\leq v_F(0).$$

Since  $1 \neq w_F(0) \not\leq v_F(0)$ , by (H), exists  $c \in A$  such that

$$c \not\leq w_F(0) \quad \text{y} \quad \neg c \not\leq v_F(0).$$

If  $\neg c \not\leq v_F(0)$ , then  $c \notin F$  □

## 9. MARCOS KC

**Definition 9.1.** Let  $A$  be a frame. We say that  $A$  is:

- (1) *KC* if each compact quotient is closed.
- (2) *Hausdorff closed compact* (or *KCH* for short) if each compact quotient of  $A$  is closed and Hausdorff.

La Definición 9.1 nos dice que si  $j \in NA$ , entonces  $\nabla(j) \in A^\wedge$  y  $j = u_a$  para algún  $a \in A$ . De manera adicional, para 2) pedimos que  $A_j$  cumpla la propiedad (H).

**Proposition 9.2.** *If  $A \in \text{Frm}$  satisfies KC, then  $A_j$  satisfies KC for each  $j \in N(A)$ .*

*Proof.* Consider  $k \in NA_j$  such that  $(A_j)_k$  is compact. If  $(A_j)_k$  is compact, then  $\nabla(k) \in A_j^\wedge$  and, by Proposition 3.1,  $j_* \nabla(k) \in A^\wedge$ .

Let  $l = j_* \circ k \circ j^* \in NA$  be, then  $A_l$  is a compact quotient of  $A$  and there exists  $a \in A$  such that  $l = u_a$ . Thus

$$\begin{array}{ccccccc} & & & & l & & \\ & & & & \curvearrowright & & \\ A & \xrightarrow{j^*} & A_j & \xrightarrow{k} & (A_j)_k & \xrightarrow{j_*} & A_j \subseteq A \end{array}$$

and  $a \vee x = k(j(x))$ . Therefore, if  $x = a$ ,  $k(j(x)) = a$ .

We need that  $k = u_b$  for some  $b \in A_j$ . If  $x \in A_j$  and  $b = j(a)$

$$\begin{aligned} u_b(x) &= b \vee x = b \vee j(x) = j(j(a) \vee j(x)) \\ &= j(k(j(a)) \vee x) \\ &= j(u_a(x)) \\ &= j(k(x)) \\ &= k(x). \end{aligned}$$

Hence  $u_b = k$ . □

**Proposition 9.3.** *If  $A$  is a  $KC$  frame, then  $A$  is a  $T_1$  frame.*

*Proof.* Let  $p \in \text{pt } A$  and  $a \in A$  such that  $p \leq a \leq 1$ . Consider

$$w_p(x) = \begin{cases} 1 & \text{si } x \not\leq p \\ p & \text{si } x \leq p \end{cases}$$

to  $x \in A$ .  $P = \nabla(w_p) = \{x \in A \mid x \not\leq p\}$  is a completely prime filter (in particular,  $P \in A^\wedge$ ). Because  $A$  is  $KC$ , then  $A_{w_p}$  is a compact closed quotient. Thus  $u_p = w_p$  and

$$u_p(a) = a \quad \text{y} \quad w_p(a) = 1.$$

that is,  $a = 1$ . Therefore  $p$  is maximal. □

One of our aims is study nuclei such that produce compact quotient, particularly, nuclei that produce compact closed quotient.

Define the following subset of  $A$

$$uA = \{a \in A \mid A_{u_a} \text{ es un cociente compacto}\}.$$

Note that, to  $a \in uA$  we have  $\nabla(u_a) \in A^\wedge$ , that is, define the map

$$\alpha: uA \rightarrow A^\wedge.$$

Let us see the properties of this map.

**Lemma 9.4.** *For  $uA \subseteq A$  as before, the following statements hold:*

- (1) *If  $A$  is tidy, then  $\alpha$  is surjective.*
- (2) *If  $u_a$  is fitted for all  $a \in uA$ , then  $\alpha$  is injective.*
- (3) *If  $A$  is tidy and  $a \in uA$ , then  $\alpha$  is bijective if and only if  $u_a$  is fitted.*

*Proof.*

- (1) Consider  $F = \nabla(j) \in A^\wedge$  and  $v_F$  the fitted nucleus associated with  $F$ . If  $A$  is tidy, then  $v_F = u_d$ , where  $d = v_F(0) \in A$ . Hence,

$$\alpha(d) = \nabla(u_d) = \nabla(v_F) = F.$$

and  $\alpha$  is surjective.

- (2) Consider  $a, b \in uA$  such that  $\alpha(a) = \alpha(b)$ , that is,  $\nabla(u_a) = \nabla(u_b)$ . By hypothesis,  $u_a, u_b$  are fitted nuclei, that is,

$$u_c \leq u_d \quad \text{y} \quad u_c \geq u_d.$$

Evaluating both nuclei at 0 we get  $c = d$  and therefore  $\alpha$  is injective.

- (3) Let  $A$  be tidy,  $a \in uA$  and suppose that  $\alpha$  is bijective. Consider  $F = \nabla(u_a)$  and  $v_F$  the fitted nucleus associated with  $F$ . By efficiency,  $v_F = u_d$ , that is,

$$\alpha(a) = \nabla(u_a) = \nabla(v_F) = \nabla(u_d) = \alpha(d)$$

where  $d = v_F(0)$ .

By hypothesis,  $\alpha$  is bijective, then, by injectivity,  $a = d$ . Therefore  $v_F = u_a$  and  $u_a$  is fitted.

Conversely, since  $A$  is tidy, then by 1),  $\alpha$  is surjective. By hypothesis,  $u_a$  is fitted and, by 2),  $\alpha$  is injective. Therefore  $\alpha$  is bijective. □

Using  $KC$  frames, we have the following result.

**Corollary 9.5.** *Let  $A$  be a  $KC$  frame and  $F \in A^\wedge$ . Then  $\alpha$  is bijective if and only if for all  $j \in [v_F, w_F]$ ,  $j$  is fitted.*

*Proof.* Suppose that  $A$  is a  $KC$  frame and let  $F \in A^\wedge$ . By definition, for all  $j \in [v_F, w_F]$ , we have  $j = u_a$  for some  $a \in A$ . Since  $A$  is a  $KC$  frame, it is efficient and by the previous lemma,  $\alpha$  is bijective if and only if for all  $j \in [v_F, w_F]$ ,  $j$  is fitted. □

**Proposition 9.6.** *Let  $A$  be a frame.  $A$  is  $KC$  if and only if  $A$  is tidy and  $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ .*

*Proof.*

$\Rightarrow$ ): Suppose that  $A$  is a  $KC$  frame. Trivially,  $KC$  implies efficient and thus, it remains to verify that  $KC$  implies that  $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$  or equivalently, by properties of the right adjoint,

$$v_F = \mathcal{U}_* \circ v_\nabla \circ \mathcal{U}.$$

If  $A$  is a  $KC$  frame, then for any  $F \in A^\wedge$ , it holds that for all  $j \in [v_F, w_F]$ ,  $j = u_a$  for some  $a \in A$ . In particular,

$$v_F = u_d, \quad \text{y} \quad \mathcal{U}_* \circ v_\nabla \circ \mathcal{U} = u_a$$

where  $d, a \in A$ .

By construction, always holds that  $u_d \leq u_a$ . To see the other inequality, we verify that the function  $\alpha$  is injective. Note that  $d \leq a$

$\Leftarrow$ ): Suppose that  $A$  is efficient and  $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$  for all HM triples  $(F, Q, \nabla)$ . First, if  $A$  is efficient, then  $A$  is  $T_1$ . Moreover, if  $A$  is efficient and  $\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$ , then  $A$  is strongly stacked. Therefore,  $A$  is  $T_1$

and strongly stacked. The above implies that  $v_F = w_F$  and, by efficiency,  $v_F = u_d$ , where  $d = v_F(0)$ . Thus, for all  $j \in [v_F, w_F]$ ,  $j = u_d$ , meaning that  $A$  is  $KC$ .

□

## 10. EXAMPLES

**10.1. The frame  $[0, 1]$ .** Following the idea presented in [Sim06b] (or, alternatively, in [Sim04]), let us consider the linearly ordered frame

$$A = [0, 1].$$

In this case the following observations hold.

- (1) For all  $a, b \in A$  the implication  $(a \succ b)$  is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

- (2) For  $a, x \in A$ , the distinguished nuclei  $u_a, v_a, w_a$  are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases}, \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

- (3) If  $F \subseteq A$  is a filter, then  $F$  is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some  $a \in A$ . In particular,  $F \in A^\wedge$  if and only if  $F = (a, 1]$ .

- (4) If  $F \in A^\wedge$ , the adjusted nucleus associated  $v_F$  is calculated as

$$(16) \quad v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases}$$

where  $l_m = v_m \wedge w_m$  and  $m = \bigwedge F$ .

- (5) For  $S = \text{pt } A$  it holds that  $S = [0, 1)$  and the open sets in  $\mathcal{O}S$  are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

- (6) The frame  $A$  is not  $T_1$ , since no  $p \in S$  is maximal. Consequently,  $A$  does not satisfy any stronger separation property in  $\text{Frm}$ .

- (7) For  $F \in A^\wedge$  the compact saturated set associated  $Q \in \mathcal{Q}S$  is of the form  $Q = [0, m]$ .

- (8) It holds that  $[Q'] = v_F$ ; therefore,  $S$  is strongly stacked.

- (9) Since  $[0, x) \in \mathcal{O}S$ , then  $[x, 1] \in \mathcal{C}S$  and  $Q = [0, m]$ . Therefore, the compact subsets are not closed; that is,  $S$  is not packed.

- (10) The frame  $A$  is not efficient.

The above resume some characteristics about the frame  $[0, 1]$ . Now, we will perform the analysis of the patch constructions of  $A$  and its point space  $S$ .

Remember that for  $A \in \text{Frm}$  and  $S \in \text{Top}$  we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where  $\text{Pbase}$  generates the patch frame of  $A$  and  $\text{pbase}$  generates the topology of the patch space of  $S$ . By (16) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1]\}.$$

For  $x \in A$  we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of  $x$  and  $m$ .

**Case 1:** If  $x < m$ , then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

**Case 2:** If  $x = m$ , then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

**Case 3:** If  $x > m$ , then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of  $a$  does not affect the evaluation. Thus, the only non-trivial case occurs when  $m < x$ , where the value of  $(u_a \wedge l_m)(x)$  depends on the relative position of  $a$  and  $x$ . We therefore distinguish the following two subcases.

**Case 3.1:** If  $m < x < a$ , then

$$(u_a \wedge l_m)(x) = a.$$

**Case 3.2:** If  $m < a \leq x$ , then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of  $\text{Pbase}$  correspond precisely to the pairs  $(m, a)$  with  $m < a$ . Consequently, the frame  $PA$  is determined by the open intervals of the form  $(m, a)$ , where  $m \in \text{pt } A$  and  $a \in A$ .

We now turn to the question of whether  $PA$  satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let  $A$  be an arbitrary frame and let  $S = \text{pt } A$  be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & PA & \hookrightarrow & NA \\ \downarrow \mathcal{U}_A & & \downarrow P(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & P\mathcal{O}S & \hookrightarrow & N\mathcal{O}S \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

where  $\mathcal{O}^f S$  is the Skula topology (or front topology) of  $S$ . In [SS06b] is named *the patch assembly diagram* and more details about it can also be found in [Sex03].

In our case,  $\mathcal{U}_A$  is an isomorphism. Moreover, under  $\mathcal{U}_A$ , the patch construction turns out to be functorial and, therefore,  $PA \cong POS$ . Similarly,  $NA \cong NOS$ . To describe explicitly  $POS$  and  $NOS$ , we use the following result, which can be consulted in [ÁBMZ20].

**Proposition 10.1.** *Let  $S$  be a partially ordered set.*

- (1) *If  $S$  has no infinite antichains, then  $NOS$  is spatial.*
- (2) *If  $S$  is totally ordered, then  $NOS$  is spatial.*

Thus, by Proposition 10.1 we have that  $NOS$  is spatial and therefore  $NOS \cong \mathcal{O}^f S$ . With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong \downarrow & & \downarrow \cong & & \downarrow \cong \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame  $PA$ , we can study the patch topology  $\mathcal{O}^p S$ . Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0, a) \cap [m, 1) \mid a \in [0, 1], m \in [0, 1)\} = \{(m, a)\}.$$

Since the topology generated by the intervals  $(m, a)$  coincides with the usual topology on  $[0, 1)$ , the space  ${}^p S$  is Hausdorff. Consequently,  $\mathcal{O}^p S \cong PA$  satisfies **(H)**, even when  $A$  is not  $T_1$  nor tidy.

**Lemma 10.2.** *Let  $A$  be a frame, if the morphism  $pU: pA \rightarrow p\mathcal{O}S$  is an isomorphism, then  $A$  is strongly stacked.*

*Proof.* Consider any open filter  $F$  on  $A$  and the compact saturated determined by it, say  $Q$ , thus we have the nuclei  $v_F$  and  $U_*[Q']U = Q[U]$ , we need to observe the action of  $pU$  on these two nuclei, in order to do that remember that we have a commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & pA & \xhookrightarrow{\iota} & NA \\ \mathcal{U}_A \downarrow & & \downarrow p(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & p\mathcal{O}S & \xhookrightarrow{\iota} & NOS \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

since  $N(\mathcal{U})\iota = v\mathcal{U}$ , that is,  $p\mathcal{U}$  is the restriction of  $N(\mathcal{U})$  to  $pA$ , then

$$\text{p}\mathcal{U}(v_F) = N(\mathcal{U})(v_F) = \bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism  $\mathcal{U}$  transforms open filters we have

$$\bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee \{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus  $v_{\mathcal{U}F}$ .

□

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**Proposition 10.3.** *Si  $A$  satisface  $(\mathbf{H})$ , entonces  $A$  es apilado.*

*Proof.* Consideremos  $A$  un marco y  $(F, Q)$  un par HM. Notemos que si  $A$  satisface  $(\mathbf{H})$ , entonces  $S = \text{pt } A$  es  $T_2$  y por lo tanto

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F$$

donde  $w_F(x) = \bigwedge \{p \in Q \mid x \leq q\}$ .

Debemos verificar que, para  $s = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = \bigwedge Q$ ,

$$s \leq v_F(0).$$

A manera de contradicción, supongamos que  $s \not\leq v_F(0)$ . Dado que  $w_F \neq \text{tp}$ ,  $1 \neq s$  y, por hipótesis, existe  $c \in A$  tal que

$$1) \ c \not\leq s \quad \text{y} \quad 2) \ \neg c \not\leq v_F(0)$$

De 1), notemos que existe  $r \in Q$  tal que  $c \not\leq r$ , pues en caso contrario, si

$$\forall r \in Q \text{ tal que } c \leq r \Rightarrow c \leq \bigwedge Q = s$$

lo cual es una contradicción.

Por 2), si  $\neg c \not\leq v_F(0)$ , entonces  $c \notin F$ , pues si  $c \in F$ ,

$$\neg c = v_c(0) \leq f_F(0) \leq v_F(0)$$

lo cual sería una contradicción. Así, existe  $q \in Q$  tal que  $c \leq q$ .

Por lo tanto, de 1) y 2) obtenemos que existe  $c \in A$  y  $q, r \in Q$  tales que

$$c \leq q \quad \text{y} \quad c \not\leq r.$$

Además, por  $T_1$ ,  $q \neq r$ . Notemos que

$$r \vee q = r \vee c = 1,$$

entonces

$$\neg r \leq c \leq q \quad \text{y} \quad \neg q \leq \neg c \leq r.$$

Por HM, sabemos que para todo  $a \in F$  se satisface que

$$Q \subseteq \mathcal{U}(a) \iff [\forall p \in Q](a \not\leq p),$$

en particular  $a \not\leq q, r$ , es decir,  $a \vee q = a \vee r = 1$ . Por lo tanto,

$$\neg q, \neg r \leq a$$

para todo  $a \in F$ . Luego,  $(\neg q \vee \neg r) \leq a$  para todo  $a \in F$ .

Consideremos  $d = \neg q \vee \neg r$ . De aquí que

$$d \leq a \Rightarrow v_a \leq v_d \Rightarrow v_F \leq v_d.$$

Evaluando en 0 obtenemos que  $v_F(0) \leq \neg d$ . □